4.2. Report grading key

Title (5%)

- Unclear title not related to the goal of the experiment. A title that do not contain descriptive words not relating to the nature of the experiment or to any of its keywords (-3).
- *Extremely simplistic and non-descriptive titles (-5).

For example “Average velocity” is a bad title.

Abstract (10%)

The abstract must contain the experimental objectives, the reasons for the design of the experiment, and rationale for how the experiment was performed, significant results and brief conclusions. Deductions in grading are for the following:

The text doesn’t answer one of the following questions:

- What we want to test? (goals) (-3)
- What is being measured and how? (-3)
- What are the primary results? What are the main conclusions? (-3)

General mistakes:

- Writing using “bullet points” and not prose descriptions in paragraphs (-8)
- Extremely long explanations (for instance beyond 2 paragraphs) (-3)
- *An overly brief abstract containing insufficient detail (-5)

Theoretical background (10%)

- Missing description of equations (-3)
- Missing equations that are used later (-5)
- Missing definitions of physical quantities (-5)
- *Too much general content or irrelevant text or formulas that are not used in the experiment, observation, or in the development of the conclusions (-8)

Materials and methods (10%)

Do not over explain the use of instruments like computers, calculators or software like Excel etc. Computers and software are only tools. Keep to the physics.

- Experimental procedure just copied from guide (-10)
- Missing diagram or photo of the experimental system (-3)

- Doesn't written what was measured and how (-3)
- Missing explanation on directly measured quantities (-3)

Results and Discussion (45%)

General

- If the results section shows no systematic logic. That is if it is written with unrelated tasks and figures or edited without explanatory prose much like lab notebook data. If this section reads like a collection of unconnected and unexplained calculations. (-20)
- Too much raw data presented (lengthy calculations, regression tables, items that should be presents in the appendix appear in the results and discussion section). It interferes with understanding of the relevant information. (-10)
- Do not include long calculations and algebra derivations. Reference the equation that is evaluated and state result. Only give derivations if it is asked or you think it is crucial to understand your idea. Include these calculations in the Appendix. (-5)
- **Missing the task as specified in the lab guide. (-10)**
- **Missing curve fitting when it is required (-10)**
- **Missing results of curve fitting/ missing errors (-5)**
- Failing to use scientific writing: Incorrect spelling of quantity (proper spelling is name + value + error + units). Incorrect rounding of values. Nonexclusive use of SI units (correct-105m and μm, incorrect-inch and Gauss). (-5)
- Missing comparison between calculated values and theoretically predicted values. (-5)
- Missing conclusion after calculating the comparison criterion. (-5)
- Unfounded or false claims. (-5/-10)
- Not enough measurements to draw conclusions. (-10)

Figures/ Tables:

- Prior to displaying data, missing text explaining the purpose of the graph/table and how the measurements were taken. (-5)
- **Measured data is presented as discrete points (without connecting lines). The results of theoretical line are displayed as a continuous line. (-5)**
- Wrong format: graphs missing axes labels, number and title, legend, uncertainty bars (if required). (-5)
- Wrong format: tables missing names for rows and columns, physical units, number and title, uncertainty of measurements. (-5)
- Missing discussion after the figure/table is presented. (-5)

Conclusions (10%)

- Missing conclusions on the calculated quantities (-5)
- If agree to theory, what is the conclusion from the experiment? (-5)
- If disagree with theory, what is the explanation? (-5)
- Conclusions are not written as text but only as separate points (e.g. "bullet points") (-8)
- Too long conclusions section (-3)
- Too short conclusions section (-5)

References (+5%)

Additional 5 points will be given if you use additional sources to the guides to validate your statements and ideas. However, less bonus points will be given if

- Statements and literature information appear in the report without proper citation (-5)
- Citing irrelevant sources (-5)
- Citing in a different format then what stated in the guidelines (-2)

Appendix (10%)

- Missing error of propagation derivation (-3)
- Missing error of propagation numerical calculations (-3)
- Missing curve fitting report (-3)

Bonus (up to 15%)

The maximum score for the lab report is 100. Additional grades are given for the following:

- A list of relevant scientific sources, written in APA format, with references cited and applicable to specific text in the report.
- The student performs additional measurements or further data processing and presents them in the report in a manner relevant to the experiment.
- The student finds an insightful physical explanation (e.g. beyond the theory presented) and explains the results received (if not conforming to the proposed theory).